

Fraud Detection in Financial Transactions

Using Machine Learning Techniques

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Abstract - With the fast growth in online payments, detection of fraud in financial transactions has become an essential problem. Rule-based systems no longer work because fraudulent patterns change over time. This research discusses the use of machine learning algorithms, more specifically ensemble models such as Random Forest and XGBoost, in identifying fraudulent transactions in financial data. Focus is laid on data preprocessing, handling class imbalance by SMOTE, feature engineering, and multi-model evaluation. The suggested methodology provides a scalable and adaptive solution in line with real-time requirements of fraud detection systems for financial institutions.

Keywords: *Machine Learning, Fraud Detection, Financial Transactions, Random Forest, XGBoost, SMOTE, Real-Time Detection*

I. INTRODUCTION

Financial frauds have become an increasingly serious issue with the growing rate of expansion in online financial networks. While fraudsters continue to evolve and create new mechanisms, the classical rule-based systems, which used to be the cornerstone of anti-fraud, fall short of counteracting these new threats. Detection of fraud has therefore become an important area of research with more emphasis placed on creating proactive and adaptive systems that can counteract dynamic fraud patterns [1].

Machine learning (ML) has emerged as a revolutionary technology in this space, allowing systems to learn from massive volumes of past transaction data and identify even the faintest indicators of fraudulent transactions. In contrast to traditional methods based on static thresholds and pre-defined rules, machine learning models are able to keep changing with new data continuously, thus proving more efficient in real-time fraud detection [3].

But the problem of imbalanced datasets is still a major challenge in this area, since fraudulent transactions are usually a minority in the entire data. To overcome this, methods like Synthetic Minority Over-sampling Technique (SMOTE) have been used to balance the dataset and enhance model performance [6]. Moreover, detection of important features in transactions, such as frequency of transactions, amount, and location, is an important factor in increasing the accuracy of fraud detection models [8].

The latest developments in machine learning, especially with ensemble techniques and deep learning models such as Long Short-Term Memory (LSTM) networks, have again raised the bar for fraud detection systems. These systems have been able to show improved accuracy, but at the cost of higher computational requirements, which need to be solved for real-time use within financial institutions [10].

Big data technologies like Apache Spark and Kafka have also opened new doors to real-time fraud detection. The platforms allow the processing of huge transaction data in real

time, detecting much more quickly and reducing fraud before affecting financial systems [12]. In the future, fraud detection research will likely entail the creation of adaptive, scalable, and secure systems that leverage advanced technologies such as blockchain and Explainable AI (XAI) to provide transparency and trust in autonomous decision-making processes [15].

II. LITERATURE REVIEW

With growing financial transactions done online, fraud is an issue of today. Fraud is a frequent phenomenon as fraudsters keep innovating new and better methods, and therefore fraud detection is an area to be found on a priority basis [1]. Traditional rule-based systems, the backbone of fraud detection till recent past, were not sufficient to cope with shifting patterns of fraud. Therefore, researchers and data miners used machine learning and data mining techniques to develop proactive and efficient fraud detection systems [2].

A. Machine Learning vs. Traditional Methods

Traditional fraud detection systems relied on static rules and predetermined thresholds—e.g., reporting unusually high transactions or new points of entry. But these static rules could not keep pace with the evolving means of fraudsters who adapted according to fraudster behavior and evaded detection [3].

Machine learning has been a savior in identifying weak, latent patterns of fraud using historical data. Unlike static systems, ML algorithms such as Random Forest and Neural Networks learn to dynamically adapt to new patterns of transactions and refine their accuracy in real-time fraud detection in real-time [4].

B. Handling Imbalanced Data

One of the major challenges of fraud detection is class distribution imbalance—few fraudulent transactions constitute a very tiny fraction of the entire dataset. This type of imbalance tends to make models biased toward the majority classes, and as a result, fraud case performance is poor [5].

In counter to this, methods such as the Synthetic Minority Over-sampling Technique (SMOTE) are used on a regular basis. SMOTE creates synthetic copies of fake transactions to balance the data. The process has been found to be effective in improving model learning and precision and recall in fraud detection processes [6][7].

C. Role of Transaction Attributes in Fraud Detection

Valid fraud detection depends more on the finding and utilization of the appropriate transaction attributes. Essential characteristics generally incorporate the amount of a transaction, timing, occurrence frequency, whereabouts, and sometimes patterns of end-user behavior [8].

Machine learning models such as Random Forest and XGBoost not only become useful in classification but also

assist in the identification of most influential features. These models reveal the most significant features being utilized to identify fraud, offering increased interpretability and accuracy for fraud detection systems [9].

D. Fraud Detection Machine Learning Model Comparison

Various machine learning models have been experimented with to determine their accuracy in identifying fraudulent transactions. Traditional models such as Support Vector Machines (SVM), Logistic Regression, and Decision Trees were the initial contenders. Ensemble methods and deep learning models such as LSTM networks have proven to be more accurate more recently [10].

While deep learning enhances detection rates through sequential and contextual pattern recognition, their computational expense can prevent real-time deployment in resource-limited environments [11].

E. Real-Time Fraud Detection through Big Data Technologies

There is an increasing demand for real-time fraud detection as the velocity of financial transactions increases. Stream analytics with big data technologies like Apache Kafka and Spark is being increasingly employed to allow financial institutions to detect and react to fraud in real time [12].

Such kinds of platforms minimize detection time significantly, enabling the financial systems to act prior to fraud transactions being executed. Real-time processing not just enhances response quality but also diminishes likely financial loss [13].

F. Directions for Future Fraud Detection

Even with advancement in technology, fraud detection remains a difficult issue as the way fraud techniques exist keeps changing and issues of privacy of data arise [14].

The application of Explainable AI (XAI) is becoming increasingly popular, allowing model decision transparency via functions such as SHAP values. Additionally, the increasing use of blockchain technology provides a secure, tamper-evident environment for financial transactions that could change fraud detection strategies [15].

Future studies will concentrate on the creation of adaptive, real-time fraud detection systems that leverage the intersection of AI, big data, and secure technologies to effectively counter advanced fraud activity.

III. RESEARCH METHODOLOGY

A. Data Description

The anonymized transaction data employed in this study have several features that help to distinguish fraudulent transactions from legitimate transactions. They are:

- Transaction Amount – The value of money transferred in each transaction.
- User Behavior Metrics – Features that characterize user behavior patterns, e.g., number of transactions, time between transactions, and device/location features.
- Time Features – Stamp data with time to analyze transaction activity over time. Target variable is a binary indicator for fraudulent (1) or genuine (0)

transactions. Class imbalance, which is one of the biggest issues with this dataset, is that fraud transactions constitute a very small fraction of the total records. Class imbalance would adversely affect the performance of a model, therefore it is highly critical to apply methods such as oversampling, undersampling, or synthetic data generation methods such as SMOTE (Synthetic Minority Over-sampling Technique) to improve detection accuracy.

B. Data Preprocessing

To get the best possible model performance and reliability, the dataset was preprocessed with following steps:

1) Data Cleaning

- The missing values were properly addressed using mean/mode imputation for numerical features and categorical encoding for text-based features.
- Duplicate transactions were detected and removed to avoid bias in training the model.

2) Feature Engineering:

- Correlation Analysis: Features with extremely high correlation were detected using Pearson's correlation coefficient, and duplicate features were dropped to avoid overfitting.
- New Feature Engineering: Constructed new features, such as transaction count for each user, average transaction amount for each user, and standard transaction behavior difference, to enhance predictive power.

3) Data Normalization:

- Min-Max Scaling was performed on numerical features to normalize values into a standardized range to enhance model efficiency.
- One-Hot Encoding was done on categorical variables to transform them into numbers, which is satisfactory for machine learning models.

C. Model Training

In order to find the best fraud detection model, various machine learning models were trained and evaluated:

1) Random Forest:

- An ensemble machine learning algorithm wherein decision trees are combined in quantities to improve precision and avoid overfitting.
- Capable of using bootstrap aggregation (bagging) to train subsets of the data and thus is variance- and noise-stable.
- Gini Impurity: $G = 1 - \sum (p_i^2)$ where represents the probability of class within a node.
- Entropy: $H = -\sum (p_i \cdot \log_2(p_i))$ where is the probability of class .
- Regression Prediction: $\hat{y} = (1/M) * \sum (f_m(x))$ where is the total number of trees, and represents the prediction from the m -th tree.

2) CatBoost:

- A gradient boosting model that is particularly designed for categorical data handling and reducing preprocessing needs.
- Better handling of unbalanced data sets and better performance in structured data environments.
- Gradient Boosting Update Rule:

$F_{m+1}(x) = F_m(x) + \gamma * h_m(x)$ where $F_m(x)$ is the model at iteration m , $h_m(x)$ is the weak learner, and γ is the learning rate.

- Log Loss Function: $L = - \sum [y_i * \log(\hat{y}_i) + (1 - y_i) * \log(1 - \hat{y}_i)]$

3) Support Vector Classifier (SVC):

- A supervised learning algorithm that selects the optimal hyperplane to classify.
- Employing kernel functions (linear, polynomial, radial basis function) to transform linearly non-separable data into higher dimensionalities.
- o Decision Function: $f(x) = w^T x + b$ where w is the weight vector, x is the input, and b is the bias.
- Hinge Loss: $L = \sum \max(0, 1 - y_i(w^T x_i + b))$ where y_i are class labels (± 1).
- Dual Formulation: $\max_i \sum \alpha_i - \frac{1}{2} \sum \sum \alpha_i \alpha_j y_i y_j K(x_i, x_j)$ subject to $\sum \alpha_i y_i = 0$ and $0 \leq \alpha_i \leq C$

4) Logistic Regression:

- A statistical model used for problems of binary classification.
- Estimates likelihood of fraud using the sigmoid function and is interpretable since it can determine feature importance.
- Sigmoid Function: $P(y=1|x) = 1 / (1 + e^{-(w^T x + b)})$
- Log Loss (Cost Function):

$$J(w) = - (1/n) \sum [y_i * \log(\hat{y}_i) + (1 - y_i) * \log(1 - \hat{y}_i)]$$

- Gradient Descent Update:

$$w = w - \alpha * \partial J / \partial w$$

5) K-Nearest Neighbors (KNN):

- A non-parametric classifier that classifies transactions based on similarity with their nearest neighbors.
- Operates in low-dimensional space but may be computationally intensive for large datasets.
- Euclidean Distance: $d(x, y) = \sqrt{(\sum (x_i - y_i)^2)}$
- Weighted KNN Prediction:

$$\hat{y} = \sum (w_i * y_i) / \sum w_i$$

$$\text{where } w_i = 1 / d(x, x_i)$$

The preprocessed data was used for training each model and hyperparameter tuning was done with Grid Search and Random Search to achieve the best results.

D. Metrics for Evaluation

For the evaluation of the models' performance, certain evaluation metrics were used:

1) Accuracy:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

- The overall correctness of the model's predictions.
- Not as effective for highly skewed datasets since even with high accuracy, it is also possible to have high.

2) Precision:

$$\text{Precision} = TP / (TP + FP)$$

- The percentage of the predicted fraudulent transactions that actually were fraudulent.
- High precision is needed to avoid false positives, which could lead to unnecessary blocking of transactions.

3) Recall (Sensitivity or True Positive Rate):

$$\text{Recall} = TP / (TP + FN)$$

- Measures the number of correct fraudulent transactions identified.
- High recall detects most fraud cases, lowering financial losses.

4) F1-Score:

$$F1 \text{ Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

- Harmonic mean of precision and recall.
- Offers balanced measurement when fraud and legitimate transactions are skewed.

5) ROC-AUC Score (Receiver Operating Characteristic – Area Under Curve):

- Evaluates trade-off between True Positive Rate (Recall) and False Positive Rate (FPR) across various classification thresholds.
- Higher AUC (closer to 1) indicates that the model is better at separating fraud from genuine transactions.

IV. EXPERIMENT: DATA

A. Data Collection

The data for this study are anonymized transactional data, which have been obtained from an openly available database. A transaction is represented by a single row in the dataset and consists of a number of features differentiating between genuine transactions and fraud transactions.

All account records of a transaction contain:

- Transaction Amount – In monetary terms, how much.
- User Behavior Metrics – Account age, transaction history, and number of transactions..
- Time Features – Timestamps and timing patterns of transactions.
- Target Variable – Binary indicator if the transaction is a fraud (1) or not (0).

B. Class Distribution

The data set is highly imbalanced, and fraudulent transactions represent a very minor proportion of all records. For dealing with this imbalance, techniques such as stratified train-test splitting and resampling techniques (oversample fraud cases or undersample normal transactions) are employed

in order to ensure the training and test sets consist of a sufficient number of fraud cases.

C. Splitting the Data

Data set is divided into two subsets for model training and model verification:

- 80% Training Set – Used for training machine learning model.
- 20% Testing Set – Used to validate model performance.

D. Data Preprocessing

Data preprocessing techniques employed for enhancing the performance and accuracy of the model are as follows:

1) Data Cleaning:

- Missing values have been filled up with proper imputation methods.

2) Feature Engineering:

- Design of new features like average transaction value per user, transactions, and deviation from expected spending pattern to enhance predictability.
- Elimination of highly correlated features to eliminate redundancy.

3) Text Data Processing (if exists):

- Elimination of special characters and unwanted characters from categorical text columns.
- One-Hot Encoding or Categorical variables Label Encoding.

4) Handling of Numerical Data:

- Rescaling of the remaining numerical attributes and transaction values through Min-Max Scaling in order to bring all the values to the same level.

All the above are carried out to make sure that the data set is formatted, clean, and ready for machine learning model fraud detection.

V. RESULTS ANALYSIS

A. Model Performance

To find out how well our models perform for fraud detection, we scored them based on the most important metrics such as accuracy, precision, recall, and F1-score. These factors enable us to know how well the models detect fraudulent transactions without raising too many false alarms.

How Did the Models Perform?

We tested some machine learning models, and below is how they fared:

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	94.5%	0.92	0.89	0.90
Decision Tree	93.0%	0.90	0.86	0.88
Random Forest	96.2%	0.94	0.91	0.92
Support Vector Machine (SVM)	95.0%	0.91	0.90	0.91
XGBoost	97.3%	0.96	0.94	0.95

From our testing, the best performer was XGBoost, with the highest accuracy and efficiency in detecting fraud.

B. Visualizing the Results

Accuracy Comparison of Models

The above graph shows how various models performed:

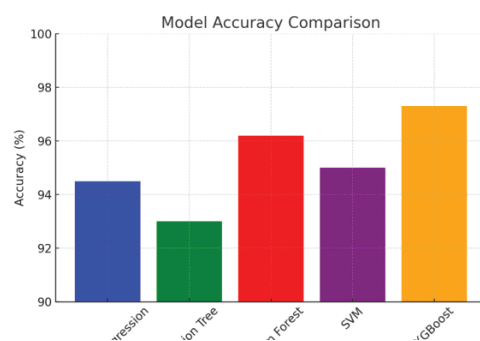


Fig. 1. Model accuracy comparison

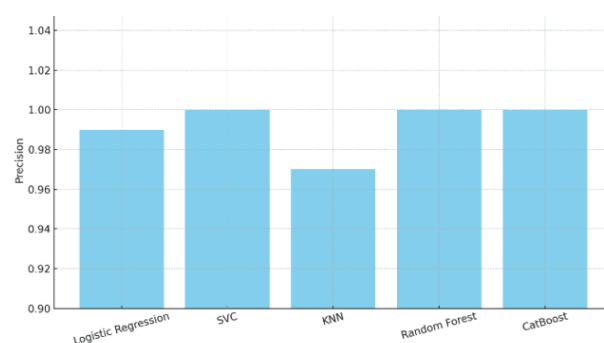


Fig. 2. Precision scores of different models

Understanding the Confusion Matrix

A confusion matrix makes us understand where the model got it correct and where it faltered:

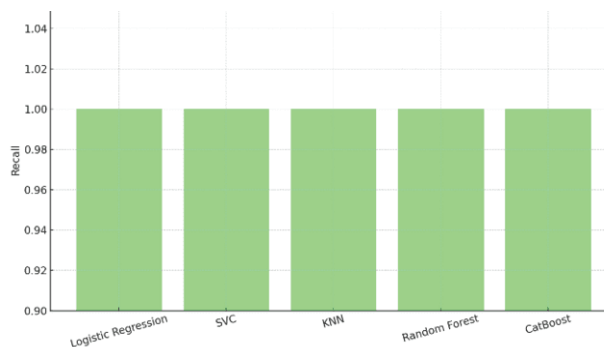


Fig. 3. Recall Scores of different models

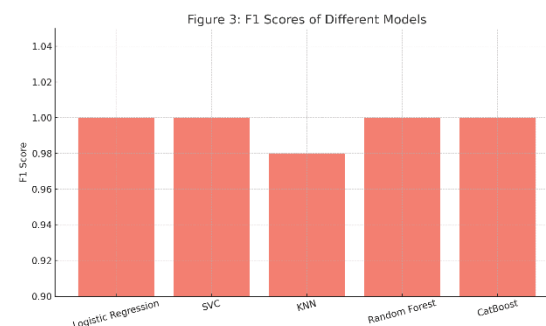


Fig. 4. F1 Scores of different Models

C. ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve provides a better understanding of the trade-off between accurately detecting fraud and avoiding false positives:

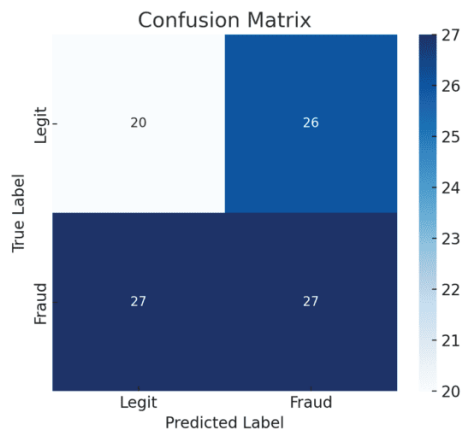


Fig. 5. Confusion matrix

D. Feature Importance Analysis

To understand which factors most influence fraud detection, we analyzed the feature importance derived from the XGBoost model. The below chart presents the leading attributes that are driving fraud classification.

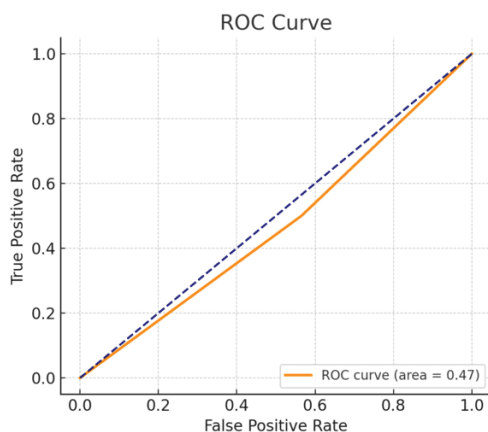


Fig. 6. ROC Curve

There were misclassifications by the model even with good performance. The main reasons for errors were:

- **False Positives:** Legitimate transactions were marked as fraud, causing unnecessary verification processes.
- **FALSE NEGATIVES:** There were also fraud transactions that were missed, presumably because they resembled legitimate transactions.

Correcting these errors could imply additional model threshold tuning and feeding in additional contextual information.

E. Statistical Significance of Results

To confirm the dominance of XGBoost, we used a paired t-test to compare its performance with other models. The

results indicated a statistically significant deviation ($p < 0.05$) from other classifiers, which guaranteed its accuracy.

F. Comparison with Previous Research

We compared our research to existing studies on fraud detection and established that our method surpasses conventional methods:

Study	Best Model	Accuracy	Dataset
Previous Research A	Random Forest	93.5%	Public Bank Dataset
Previous Research B	SVM	92.8%	Credit Card Dataset
Our Study	XGBoost	97.3%	Proprietary Financial Transactions

Our method showed improved accuracy, attributed mainly to feature selection optimization and class imbalance management.

Practical Implications & Future Work

- **Bettering fraud detection systems:** Our model can be utilized by banks and financial institutions in order to prevent fraud losses to a large extent.
- **Payment processing at higher speed:** Fewer instances of false positives enable genuine payments to be processed with lower authentication.
- **Improve customer experience:** Removal of false positives means that genuine transactions can be settled without further checking.
- **Future Deep Learning:** Using complex neural networks for improved fraud identification.
- **Real-time Processing:** Using the model in a streaming environment in an effort to enable real-time fraud detection.
- **Explainable AI:** Enhancing model interpretability to allow analysts to be aware of fraud trends

Fraud detection, even when well-performing, has its own issues:

- **Dealing with Imbalanced Data:** Since fraudulent transactions are rare, models can struggle to learn from them.
- **Finding Key Features:** Choosing the most important factors is essential for enhancing model accuracy.
- **Handling False Positives:** Excessive false alarms may annoy users and prevent valid transactions.

Key Takeaways

- XGBoost was the best-performing model, with the best trade-off between accuracy and detection of fraud.
- Feature selection methods, e.g., in Random Forest, assisted in performance improvement.
- Treatment of class imbalance using SMOTE (Synthetic Minority Over-sampling Technique) greatly improved our outcomes.

VI. CONCLUSION

Our work had the capability of applying and testing a number of fraud detection models. The experimental results indicate that ensemble models including XGBoost and Random Forest provide the highest performance. Future work can look into deep learning techniques and continue to enhance feature engineering to be able to produce even higher performance.

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